# Sahaj Bhadja

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## Summary

Software Engineer with 2+ years building **consumer-facing web products**, **AI/LLM workflows**, React/TypeScript interfaces, and scalable backend services across ambiguous, cross-functional product settings.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Advanced NLP, Web Technologies, Database Systems, Analysis of Algorithms

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

## Experience

**USC Dornsife** **September 2025 – Present**
*Software Engineer* *Los Angeles, CA*
- Built Azure-backed AI education platform with normalized SQL schemas, row-level isolation, and indexed dashboards, reducing latency from 1.2 seconds to 350 milliseconds for 500+ campus users across releases.
- Reworked FastAPI LLM job orchestration with explicit states, failure isolation, and reusable status APIs, processing 1K+ AI tasks while tripling throughput for concurrent instructor-facing workflows and dashboards.

**Grad8** **June 2025 – August 2025**
*Software Engineer Intern* *Los Angeles, CA*
- Built React and Flask AI grading workflows for uploads, rubrics, history, and model selection, supporting 150+ instructors and 10K+ daily requests through product-facing dashboards and backend APIs.
- Designed multimodal LLM review services with OCR, ThreadPoolExecutor, and secure file ingestion, reducing feedback turnaround 60% while keeping batch processing reliable, observable, and user-responsive during peak submissions.

**USC AD Labs** **January 2025 – May 2025**
*ML Research Assistant* *Los Angeles, CA*
- Built a Decision Transformer for autonomous-driving offline RL, improving planning 18% in simulation while processing 100K+ trajectory logs through reproducible PyTorch evaluation pipelines for applied AI research.
- Architected a spatial-temporal encoder combining HD-map geometry, lane context, and ego trajectories with multi-head attention, improving closed-loop stability 15% across simulated baselines for planning workflows and reports.

**Aasma Technology Solutions** **May 2023 – August 2023**
*Full Stack Developer Intern - (VibrantChem)* *India*
- Developed React and Node resource-management workflows with REST APIs, WebSockets, and modular services, synchronizing 2M+ records across locations with 33% fewer conflicts during staged rollouts and reviews.
- Implemented CRM modules for authentication, resources, billing, leads, permissions, and notifications, using Jenkins CI/CD and Docker staging to cut deployment time 50% across release cycles and fixes.

**LST Capitals** **November 2022 – January 2023**
*Web Developer Intern* *India*
- Built React Native and TypeScript booking flows for time-slot selection, reservation tracking, coupon pricing, and payment-status updates across 50K+ records with sub-200-millisecond APIs for cross-platform users daily.
- Designed Figma-led architecture for 12+ integration points, aligning Flask APIs, Firebase synchronization, payment webhooks, and client requirements to reduce design errors 25% before implementation and stakeholder handoff.

## Skills

**Languages & Platforms:** Python, Java, C++, TypeScript, JavaScript, SQL, HTML, CSS, React Native, OOP, Data Structures, Algorithms
**Frontend & Consumer Surfaces:** React, Next.js, Node.js, Flask, FastAPI, REST APIs, WebSockets, Responsive UI, Browser APIs, Product Prototyping, Reusable Components, Cross-Platform Development
**AI/ML & Data:** LLMs, Transformers, PyTorch, OCR, FAISS, Semantic Search, Prompt Workflows, Model Evaluation, Offline RL, Computer Vision, Embeddings
**Cloud & Engineering:** AWS, Azure, GCP, Docker, CI/CD, Jenkins, GitHub Actions, PostgreSQL, MongoDB, MySQL, SQL Server, Unit Testing, Integration Testing, Debugging, Agile

## Projects

**Context-Aware Web Assistant** | *TypeScript, React, Browser APIs, FastAPI, LLMs, Reusable UI*
- Designed an ambient assistant prototype with React, TypeScript, browser context capture, prompt routing, reusable UI components, and FastAPI services for concise actions across active webpages and tasks.

**Big-Brain** Live | *LLaMA-2, FAISS, PostgreSQL, GCP, REST APIs*
- Built semantic document retrieval over 50K+ documents using transformer embeddings and FAISS, enabling contextual Q&A, summarization, and sub-second retrieval through optimized similarity pipelines for assistant-style workflows online.

**PiXMod** Live | *Next.js, CLIP, SAM, Stable Diffusion, MongoDB*
- Built a responsive Next.js image platform with CLIP search, SAM segmentation, Stable Diffusion editing, MongoDB storage, and Stripe billing, processing 1K+ visual assets through consumer-facing workflows online.